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RESEARCH PAPERS

Graph Neural Networks for Candidate-Job Matching: An Inductive Learning Approach

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Abstract
This work introduces a novel graph-based approach to candidate-job matching using Graph Neural Networks (GNNs). We analyzed data from 62 real-world selection processes encompassing 8360 unique candidates and 9532 applications, characterized by extreme class imbalance (95% rejection rate). Our methodology constructs purpose-built bipartite graphs for each candidate-job pair, with 14 nodes representing candidates, jobs, and their respective attributes extracted using Large Language Models. Each graph contains a minimum of 15 edges representing semantic relationships between entities, with edge weights derived from embedding similarity measures. We empirically evaluated five GNN architectures (GCN, MIGNN, GIN, GAT, GraphConv) against standard neural networks across binary and ordinal classification tasks. In binary classification, graph-based approaches consistently outperformed non-graph baselines, with GCN achieving 65.4% balanced accuracy compared to 55.0% for the MLP baseline. GNN models also demonstrated superior minority class detection, with GCN correctly identifying 48.9% of qualified candidates versus only 8.5% for MLP. Statistical analysis revealed that higher recruitment stages correlate with increased graph connectivity, validating our graph construction methodology. While all models struggled with ordinal classification, the explicit modeling of semantic relationships through graph structures enabled effective binary discrimination for candidate screening, offering a promising direction for augmenting human decision-making in recruitment processes while maintaining interpretability.

Keywords Graph neural networks · Personnel selection · Human resource management · Large language models · Recruitment

1 Introduction

Human Resource Management (HRM) is experiencing a profound transformation driven by rapid advancements in digital technologies and artificial intelligence (AI). Within HRM, the candidate-job matching (CJM) process—where recruiters match candidates to job vacancies—has emerged as a critical challenge ripe for technological innovation. Traditionally, HR recruiters manually screen hundreds of resumes, analyze job requirements, and make complex decisions about candidate suitability. This process is not only time-consuming but also susceptible to human biases [21]. The growing volume of digital applications and increasingly diverse job roles have made this manual approach increasingly difficult to maintain efficiently [33].

Recent advances in AI, particularly in Natural Language Processing (NLP) and deep learning, show promise in automating recruitment processes [23]. Current approaches typically rely on keyword matching, resume parsing, or basic similarity metrics between candidate profiles and job descriptions. However, these methods often fail to capture subtle relationships between candidate attributes and job requirements, resulting in suboptimal matches [28]. Furthermore, most existing solutions treat matching as a one-dimensional problem, overlooking the complex interplay of skills, experiences, and qualifications that define successful job placements [20].

Traditional machine learning approaches to CJM face several key limitations. They often depend on rigid, predefined features that inadequately represent the full spectrum of candidate qualifications. Additionally, these approaches

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struggle with the unstructured nature of recruitment data, including diverse resume formats, job descriptions, and assessment results. Most current systems also lack the capability to learn from historical hiring decisions, missing valuable patterns that could guide future matches [31]. Furthermore, they typically match candidates to job listings in isolation, disregarding the broader context of multiple roles and varied organizational needs [12].

1.1 Problem Statement

The CJM task constitutes a fundamental challenge in human resource management with broad economic and social implications. At its core, CJM involves determining the degree of compatibility between a candidate and job description, where optimal matching decisions maximize both organizational performance and candidate satisfaction. Beyond its conceptual simplicity, CJM presents several interrelated computational challenges:

- *Relational Complexity* The candidate-job relationship encompasses multifaceted connections between skills, experiences, qualifications, and job requirements that cannot be adequately captured through simple vector similarity or keyword matching approaches.
- *Multimodal Data Integration* Recruitment data spans structured assessments (psychometric evaluations, skills ratings) and unstructured content (resumes, job descriptions), necessitating sophisticated integration techniques that preserve semantic relationships across modalities.
- *Transparency Requirements* Unlike many machine learning applications, recruitment decisions demand high levels of interpretability and justification, requiring models that can explain their reasoning in human-understandable terms.
- *Generalization to Novel Scenarios* Practical CJM systems must generalize effectively to previously unseen candidates and job descriptions without requiring retraining, presenting a challenging inductive learning problem.

These challenges are further compounded by practical constraints, including extreme class imbalance in recruitment outcomes (as most applicants are rejected), limited availability of cross-organizational benchmark datasets, and the inherent subjectivity in assessing candidate-job fit.

1.2 Research Objectives

To address these challenges, we propose an innovative approach combining Graph Neural Networks (GNNs) and Large Language Models (LLMs) to create a more sophisticated and adaptive CJM system. Our methodology advances the field in several key aspects:

1. To develop a graph-based representation framework for CJM that explicitly models the complex relationships between candidates, jobs, and their constituent attributes through interpretable graph structures.
2. To exploit LLMs to extract rich contextual features from unstructured resumes and job descriptions, generating meaningful entity embeddings that form the foundation of our graph construction process.
3. To establish a methodology for seamlessly integrating multimodal recruitment data (structured psychometric assessments and unstructured text) within a unified graph representation.
4. To design GNN architectures specifically optimized for the CJM task, capable of effectively processing purpose-built candidate-job graphs while maintaining generalization capabilities.
5. To empirically validate the efficacy of graph-based approaches to CJM against traditional methods using real-world recruitment data with practical evaluation metrics.

1.3 Research Hypotheses

Our investigation is guided by four core hypotheses:

1. *Structural Information Hypothesis* Graph neural networks will demonstrate superior performance compared to non-graph baseline methods on CJM tasks due to their ability to exploit relationship information encoded in graph topology that is inaccessible to vector-based approaches.
2. *Inductive Learning Hypothesis* A purpose-built graph construction methodology for candidate-job pairs will enable effective generalization to new, previously unseen recruitment scenarios without requiring retraining.
3. *Edge Semantics Hypothesis* The explicit modeling of semantic relationships between candidate and job entities through weighted edges will provide discriminative features that improve classification performance beyond what entity embeddings alone can achieve.
4. *Interpretability Hypothesis* The explicit graph structure of our approach will facilitate improved explainability of matching decisions through the visualization and analysis of edge patterns.

This research bridges a critical gap between theoretical advances in graph representation learning and practical HR applications. Our formulation of CJM as an inductive graph learning problem enables the discovery of complex patterns in historical hiring decisions while ensuring robust generalization to new scenarios.

The paper organized as follows: Sect. 2 reviews related research in AI-driven recruitment and graph neural networks.

Section 3 elaborates on our data processing strategy and the generation of entity embeddings using language models. Section 4 introduces our method for graph construction and the associated theoretical framework. Section 5 provides a detailed description of the architecture and training process of the GNN model. Section 6 presents our experimental assessments and results. Section 7 describes our findings, discussing their relevance to HR practices. Finally, Sect. 8 summarizes the principal contributions of our study and potential avenues for future research.

2 Related Work

The integration of AI in personnel selection marks a transformative development in HRM practices. Early CJM approaches primarily relied on information retrieval techniques for parsing semi-structured resumes [41, 42] and implementing basic similarity metrics between candidate profiles and job requirements [17]. While these foundational methods established important baselines, they struggled to handle the inherently unstructured nature of recruitment data and failed to capture the nuanced relationships between candidates' qualifications and job requirements.

The evolution of NLP techniques has enabled increasingly sophisticated approaches to candidate assessment. A pioneering study by Faliagka et al. [9] demonstrated the potential of digital footprints in selection processes by analyzing data from social network profiles and blogs to predict person-job fit. This work was complemented by Campion et al. [5], who conducted a large-scale study showing that NLP methods could achieve strong correlations with human ratings in CV screening. The field has since progressed to leverage transformer-based models like BERT, significantly enhancing feature extraction capabilities from resumes and job descriptions [35, 37].

The recent emergence of Large Language Models (LLMs) has further revolutionized text processing capabilities in HR applications. Researchers have developed innovative frameworks for LLM-based agents that can perform CV summarization and evaluation [14]. Additionally, Vagale et al. [34] have shown how closed-source LLMs can generate personalized candidate reports and feedback. However, the use of LLMs in candidate evaluation has raised important ethical considerations and faces significant regulatory oversight [2, 22], highlighting the need for responsible AI implementation in recruitment processes.

The evolution of deep learning has sparked innovative approaches to addressing the relational complexity of CJM. A significant breakthrough came from Zhu et al. [45], who developed a CNN-based architecture that projects resumes and job postings into a shared latent space, enabling direct quantitative comparison of candidates and positions. This work was extended by Yan et al. [40], who leveraged recurrent neural networks to capture temporal dynamics in recruitment platforms, accounting for evolving preferences and market trends. Further advancing the field, Wang et al. [37] introduced a pioneering knowledge graph-based approach that models the intricate interactions between resume features and job requirements through explicit semantic relationships. While these methods represented significant progress, they often struggled with rigid, predefined structures and showed limited adaptability to the inherent noise and variability in real-world HR data.

Beyond technical limitations, researchers have identified fundamental challenges in the effective representation and analysis of HR data. König et al. [19] uncovered a critical disconnect between computer science methodologies and established personnel selection practices, emphasizing that many AI solutions lack proper grounding in industrial-organizational psychology principles. These theoretical gaps are further exacerbated by practical concerns regarding bias, transparency, and accountability in AI-based hiring tools, as thoroughly documented by Raghavan et al. [30]. Together, these challenges underscore the need for more holistic, ethically-aware approaches.

Given the inherently relational nature of recruitment data, Graph Neural Networks (GNNs) [26, 32] emerge as a particularly promising approach. These architectures have demonstrated exceptional capabilities in modeling complex relational data across various domains, from molecular chemistry [15] and social network analysis [4] to epidemiology [27]. Despite their success in these fields, their application to HR analytics, particularly in CJM, remains largely unexplored. Current graph-based approaches in HR, exemplified by skill graphs [29], have primarily focused on static ontological relationships, overlooking the rich potential of learning dynamic patterns from historical hiring decisions and candidate-job interactions.

Our research addresses several critical gaps in the existing literature. Where previous approaches typically treat candidates and jobs as discrete entities, we propose an inductive graph learning framework that captures the fundamental interconnectedness of the recruitment ecosystem while preserving the ability to generalize to new scenarios. This methodology transcends the limitations of conventional systems' rigid structures, offering a more adaptable solution for dynamic job markets. Moreover, we successfully bridge the gap between advanced AI capabilities and practical interpretability by combining state-of-the-art LLMs for feature extraction with transparent, interpretable graph structures-a crucial advancement for HR applications where model explainability often carries equal weight with predictive accuracy.

A key innovation of our approach lies in its comprehensive data integration strategy, which addresses a fundamental limitation in existing methods. While current approaches typically handle either structured or unstructured data in isolation, our graph construction methodology seamlessly unifies standardized assessment metrics with rich textual information extracted from resumes and job descriptions. This unified representation enables a more holistic view of the recruitment process, capturing both quantifiable skills and nuanced qualitative aspects of candidate-job fit. Through extensive validation of real-world recruitment data, we demonstrate that graph-based approaches can effectively augment human decision-making while maintaining high standards of transparency and interpretability. This bridges the persistent gap between AI advances and practical HR applications, offering a more robust and adaptable solution for modern recruitment challenges.

Table 1 briefly summarizes the methods, data sources, and outcomes of related studies compared to our proposed approach. For a more detailed dissertation on the current state-of-the-art and comparison of methods, refer to Laumer et al. [23].

3 Problem Description and Data Processing

The CJM problem represents a core challenge in HRM, encompassing the intricate process of matching candidates' credentials, competencies, and qualities with job specifications and the organization's demands. Moreover, it involves computational difficulties due to the heterogeneous and predominantly unstructured nature of HR data and the inherent subjectivity in recruitment assessments. Additionally, empirical research in this domain faces methodological constraints stemming from limited public benchmark datasets and insufficient cross-disciplinary collaboration between computational scientists and HR domain experts.

This research is grounded in empirical recruitment data collected through an industrial partnership. This collaboration facilitated access to a comprehensive dataset of real-world recruitment processes, enabling the development and validation of models against authentic selection scenarios rather than synthetic approximations.

3.1 Dataset Collection and Characteristics

The dataset encompasses longitudinal recruitment data collected from January 2021 to May 2023, comprising $S = 62$ distinct selection processes with corresponding Job Descriptions (JDs). It contains data from $C = 8360$ unique candidates, resulting in $N = 9532$ candidate-application pairs due to recursive applications or recruiter-initiated matches. The selection processes exhibited substantial variability in applicant volume, with a mean of $159_{\pm 103}$ applicants per position, ranging from 26 to 648 applications, reflecting the heterogeneous nature of labor market dynamics across different roles and sectors.

In compliance with data protection regulations and ethical research practices, we implemented rigorous anonymization protocols. All personally identifiable information was systematically removed through automated preprocessing pipelines. Participants provided explicit informed consent for data collection and subsequent analytical usage in accordance with GDPR principles. The dataset representation maintains analytical utility while preserving candidate privacy through aggregate-level analysis and feature abstraction.

Each candidate record in the dataset consists of two primary components:

- A Curriculum Vitae (CV) containing semi-structured text documenting qualifications, professional trajectory, and domain-specific competencies.
- Psychometric assessment results from a proprietary questionnaire yielding 18 numerical traits, quantifying behavioral tendencies and organizational compatibility metrics.

The psychometric assessment was completed by 3895 candidates (46.6% of the total population). For analytical consistency, their trait vectors were normalized using z-score

Table 1 Summary of related works and contributions of our approach

Related work	Data	Outcome
IR and similarity-based approaches [17, 41, 42]	Semi-structured resumes	Basic CJM
Digital footprint and early NLP [5, 9]	Social profiles, CVs	Person-job fit prediction
Transformer-based models (e.g., BERT) [35, 37]	CVs, JDs	Enhanced feature extraction
Deep learning (CNN/knowledge graphs) [37, 40, 45]	CVs, job postings	Quantitative comparison, temporal dynamics
Graph-based HR analytics [29]	Skill data	Static ontological relationships
Our approach (LLM and GNN)	CVs, JDs, Psy. assessments	Graph-based (CJM)

standardization to ensure commensurability across different psychometric dimensions. For the remaining 4465 candidates, questionnaire feature vectors were initialized with null values (zero-vectors), representing the absence of psychometric data. Additionally, each JD incorporates minimum threshold values for specific psychometric dimensions, established by HR domain experts based on role-specific competency frameworks and organizational requirements.

The target variable, encoding recruitment outcomes, follows a categorical ordinal scale reflecting the hierarchical selection stages:

- `Rejected` (0): Candidates whose applications were not selected for further consideration;
- `Screened/Interviewed` (1): Candidates who advanced beyond a manual screening;
- `Shortlisted` (2): Candidates selected for final consideration;
- `Hired` (3): Candidates who received and accepted a formal employment offer.

As quantified in Table 2, the dataset exhibits pronounced class imbalance characteristic of real-world selection processes, with approximately 95% of applications resulting in rejection. This imbalance constitutes a significant challenge for supervised learning approaches, necessitating specialized algorithmic strategies and evaluation metrics.

3.2 Data Preprocessing and Feature Extraction

The preprocessing pipeline addresses multiple modality-specific challenges inherent in recruitment data. CVs, typically encoded as semi-structured documents in PDF or DOCX formats, present significant extraction challenges due to their heterogeneous layouts, embedded non-textual elements, and non-standardized structural conventions. While DOCX documents permit relatively straightforward content extraction via XML parsing, PDFs necessitate more sophisticated processing due to their presentation-oriented rather than semantically-structured nature. We employed the `PyPDF` library with custom extraction heuristics for document parsing, successfully processing all but 362 documents (4.3%) that exhibited encryption or rendering anomalies and were consequently excluded from the analytical corpus.

Table 2 Overall and grouped candidate label distribution and statistics (in %) for $S = 62$ selected job openings

Label y	Total	Grouped by S			
		Mean	Min	Max	Std
0	95.10	94.50	69.23	99.22	4.77
1	3.40	4.32	0.39	30.77	4.75
2	0.84	1.65	0.30	4.17	0.97
3	0.66	0.94	0.27	3.70	0.74

The 'Total' column represents the distribution across all job openings, while the 'Grouped by S' columns show statistics for the distribution within each selected job opening

3.2.1 Text Normalization

To ensure consistency and enhance feature extraction fidelity, the extracted text undergoes a multi-stage normalization process. First, we removed redundant whitespace, non-ASCII characters, and control sequences. Second, we applied systematic case normalization and punctuation harmonization to reduce superficial textual variability. Third, we implemented deterministic identification and removal of personally identifiable information (PII) through regular expression pattern matching and named entity recognition techniques, targeting name tokens, contact information, geographical identifiers, and demographic markers.

This normalization process transformed the heterogeneous document corpora into standardized text representations suitable for subsequent semantic analysis and feature extraction while preserving the informational content relevant to qualification assessment.

3.3 Feature Selection

The feature selection methodology was informed by theoretical constructs from industrial-organizational psychology and empirically validated recruitment practices. The selected features encapsulate both explicit credentials and latent abilities that influence candidate suitability and performance potential. Each feature category was validated through consultation with HR subject matter experts and represents distinct dimensions of the candidate-job fit paradigm.

Soft Skills represent behavioral and interpersonal competencies that facilitate workplace effectiveness beyond technical expertise. These attributes encompass emotional intelligence, adaptability quotients, conflict resolution aptitude, and collaborative tendencies that significantly influence team dynamics and organizational integration. *Hard Skills* denote quantifiable technical competencies and domain-specific knowledge measurable through standardized assessments or credential verification. These encompass programming proficiencies, technological platform expertise, analytical methodologies, and linguistic capabilities.

Education credentials function as proxy indicators for foundational knowledge acquisition and learning capacity. These formalized qualifications include academic degrees, professional certifications, and specialized training completions, providing standardized benchmarks for comparing candidates. Similarly, the *Field of Education* represents the candidate's disciplinary specialization, indicating specific knowledge domains and theoretical frameworks that inform their professional approach.

The *Industry Sector* feature captures domain-specific experiential knowledge accumulated through professional engagement in particular market segments or organizational contexts. Finally, the *Role* dimension encodes the candidate's historical position classifications and responsibility levels, providing insights into their career trajectory, managerial experience, and organizational positioning.

Collectively, these feature categories provide a comprehensive representational framework for candidate profiles. This taxonomic approach enables nuanced matching against job requirements while preserving interpretability for human decision-makers in the recruitment process.

3.3.1 Entity Extraction

Extracting structured information from an unstructured CV and JD text represents a labor-intensive process when conducted manually, requiring domain expertise and significant time investment. To address this operational challenge, we leveraged state-of-the-art Large Language Models, specifically `GPT-4` [1], to perform automated entity extraction. This approach significantly reduces processing overhead while maintaining extraction consistency across large document corpora.

Using GPT-4, we extracted six key entity categories described above from both CVs and JDs. The entity category taxonomy is formally defined as

$$\begin{aligned}\mathcal{E} =\{&\textit{Soft Skills}, \textit{Hard Skills}, \textit{Education},\\ &\textit{Field of Education}, \textit{Industry Sector}, \textit{Role}\}.\end{aligned}$$

Thus, each entity can be denoted as $\epsilon \in \mathcal{E}$.

The extraction methodology employed iteratively refined prompt engineering techniques, incorporating domain-specific extraction rules and contextual parsing instructions. The prompts were developed through a human-in-the-loop refinement process incorporating feedback from HR practitioners to optimize extraction precision and recall. This approach ensured that the extracted entities maintained both semantic accuracy and domain relevance.

3.3.2 Embedding Generation

To transform the extracted textual entities into numerical representations suitable for computational processing, we employed OpenAI's `text-embedding-3-large` model. Each entity instance $\epsilon \in \mathcal{E}$ was encoded as a dense vector in $\mathbb{R}^{768}$, denoted as $\mathbf{v}_{\epsilon,i} \in \mathbb{R}^{768}$, where i indexes the specific instance within the entity category.

This encoding process transforms each CV and JD into a structured representation organized as a *set of sets* of embedding vectors, preserving the hierarchical relationship between entity categories and their instances. To maintain dimensional consistency across all document representations, missing entity categories were imputed with null vectors $\mathbf{0} \in \mathbb{R}^{768}$. An overview of the entity categories, including embedding cardinalities and representative instances, is provided in Table 3.

This feature extraction methodology yields a rich representational foundation that captures both explicit content and latent semantic relationships within recruitment documents. The integration of structured psychometric data with LLM-derived embedding representations enables our model to simultaneously consider quantitative assessments and qualitative attributes in the matching process.

4 Graph Construction Methodology

The cornerstone of our methodology lies in the formulation of purpose-built graph structures that encode the semantic relationships between candidates and job descriptions. This approach transforms unstructured textual data into graph representations suitable for GNN processing. Rather than adopting traditional vector-space approaches that treat candidate-job matching as a direct mapping problem, we propose a graph-centric framework where each candidate-job pair is represented as a distinct graph instance. This formulation naturally frames CJM as an inductive graph learning task, enabling generalization to unseen candidates

Table 3 Overview of the entity embeddings, including their cardinality and representative lexical instances for each taxonomic category

Entities $\mathcal{E}$	#Embeddings	Examples
Soft skills	6776	Campaign development, Desire to learn, Mentoring start-ups, Problem-solving mindset
Hard skills	16,406	.NET framework, CAD, Credit analysis, Data management, English (B2)
Education	6960	Bachelor's degree in Tourism, Diploma in Accounting, Erasmus course, SAP certificate
Field of education	2780	3D design, Science, Economics, Marketing, Neuroscience
Industry sector	7628	Manufacturing, Law enforcement, Pharma, Restaurant management
Role	15,749	1 st Officer, CTO, Medical Representative, Researcher, Technician

and job descriptions—a crucial capability for real-world HR applications.

4.1 Graph Structure Design

We define each CJM graph as a heterogeneous bipartite structure $G = (V, E)$, where V comprises two types of nodes: primary nodes (candidate and job description) and entity nodes derived from $\mathcal{E}$. The edge set E contains both structural connections and semantic relationships.

Formally, let v_c and v_{jd} denote the candidate and job description nodes, respectively. These primary nodes are connected by a direct edge $(v_c, v_{jd}) \in E$ and augmented with self-loops to facilitate information propagation in multi-layer GNN architectures. For each entity category $\epsilon \in \mathcal{E}$, we create corresponding entity nodes on both the candidate ($v_{\epsilon,c}$) and job ($v_{\epsilon,jd}$) sides, establishing a symmetric star topology through edges $(v_c, v_{\epsilon,c})$ and $(v_{jd}, v_{\epsilon,jd})$.

The resulting graph structure contains $|V| = 14$ nodes (two primary nodes plus six entity nodes per side) and a minimum of $|E| = 15$ edges, as illustrated in Fig. 1. This architecture ensures comprehensive capture of candidate-job relationships while maintaining a consistent structural pattern across all instances.

The rationale behind this star-like bipartite structure is grounded in the message-passing mechanism of GNNs. This design enables direct comparisons between corresponding entity categories across the candidate-job boundary during the convolution operations. Alternative graph formulations, such as a fully-connected approach, would introduce redundant paths that dilute the semantic specificity of entity relationships. The star topology optimizes information flow by ensuring that: (1) Entity embeddings are first contextualized within their respective domains (candidate or job) through connections to primary nodes, (2) cross-domain relationships are then established through the central candidate-job edge, creating focused comparison pathways, and (3) message propagation naturally mirrors the cognitive process of comparing specific qualifications to requirements through distinct assessment channels. This architecture also maintains computational efficiency while preserving interpretability, as each edge represents a semantically meaningful relationship in the recruitment context. While our current design proves effective, future work could explore alternative graph construction paradigms, particularly in light of recent advances in learnable graph structures [24, 44].

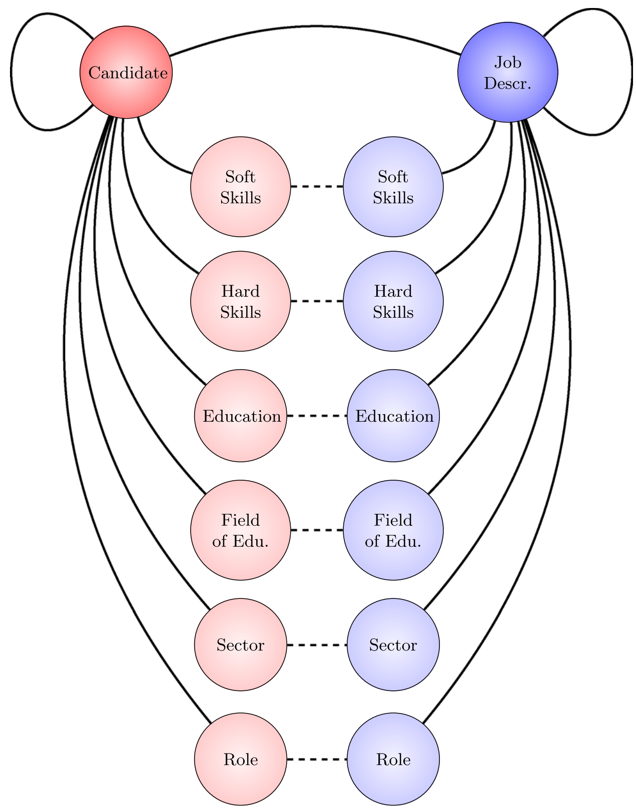


Fig. 1 Illustration of the bipartite graph structure for candidate-job matching. The graph comprises two primary nodes (candidate and job description) connected by a direct edge, augmented with self-loops. Each entity category $\epsilon \in \mathcal{E}$ is represented by entity nodes on both sides, forming a symmetric star topology

4.2 Edge Construction and Weighting

A critical contribution of our methodology lies in establishing connections between the graph's bipartite components through a principled weighting scheme. For each entity category $\epsilon \in \mathcal{E}$, we define a similarity function that quantifies the semantic proximity between corresponding entity embeddings. This function computes a normalized intersection of k-nearest neighborhoods in the embedding space, where $k = 10$:

$$\mathrm{sim}_\epsilon(i,j) = \left(\frac{\sum_{\mathbf{v}_{\epsilon,i},\mathbf{v}_{\epsilon,j}} \mathrm{kNN}(\mathbf{v}_{\epsilon,i}) \subseteq \mathrm{kNN}(\mathbf{v}_{\epsilon,j})}{|E_{\epsilon,i}| + |E_{\epsilon,j}|} \right)^{1/p} \in [0,1], \tag{1}$$

where indices i and j reference candidate and job description entities, respectively. The parameter $p = 4$ serves as a sharpening factor, amplifying the weights of highly similar entity pairs while attenuating weaker matches. This formulation effectively captures the granular similarities between technical competencies, educational credentials, and other domain-specific attributes. Figure 2 illustrates this edge construction process through two representative examples.

4.3 Node Feature Integration

The model incorporates two distinct feature representations to facilitate learning from both structural and attribute information:

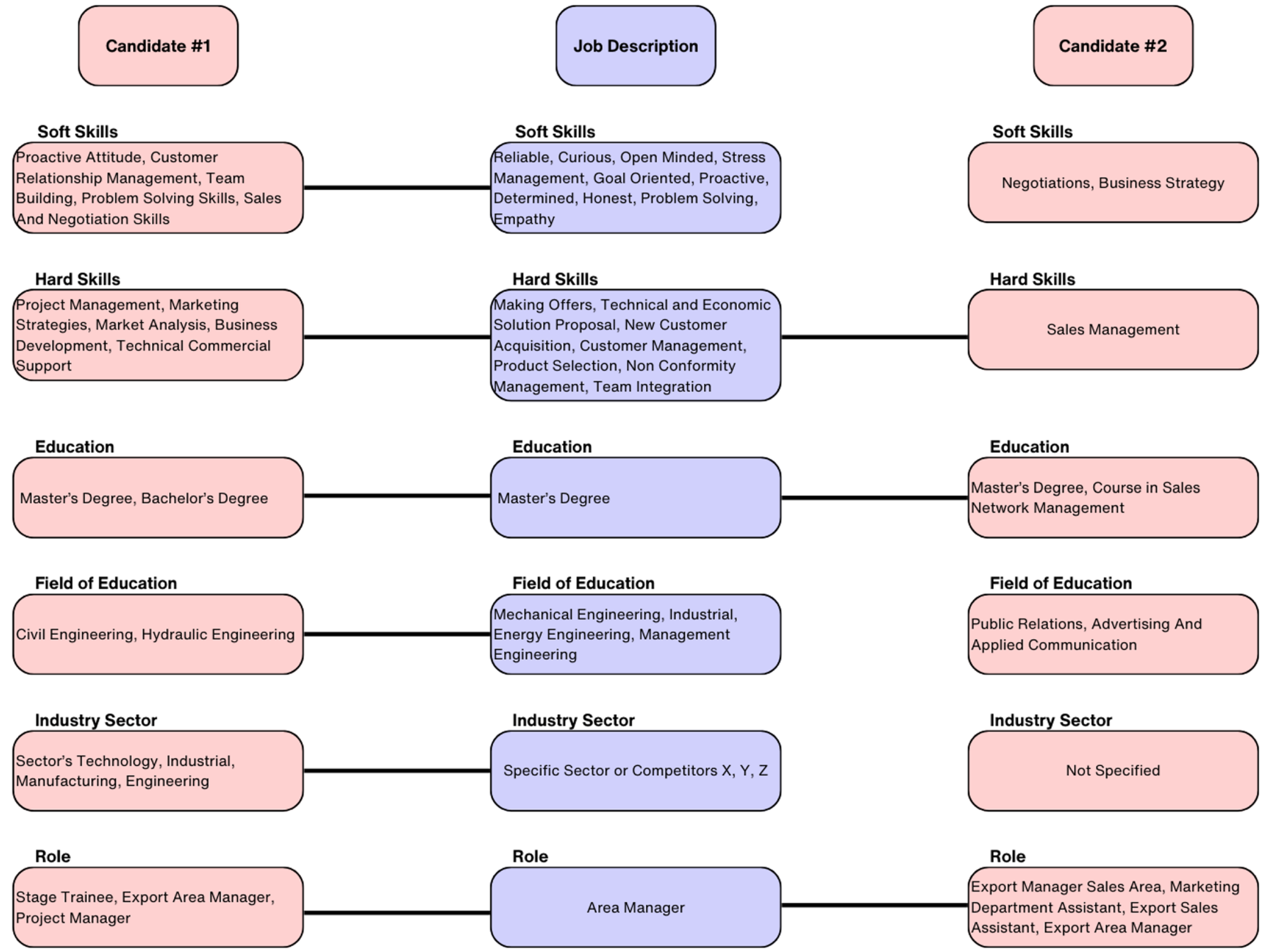


Fig. 2 Example of anonymized CJM graphs of two candidates that applied to the same position. The actual dataset is made of bipartite graphs as in Fig. 1 and with features as in Sect. 4. This example highlights the extracted entities and the resulting edges between a candidate and a JD, while all the remaining edges are omitted. Candidate #1 (left) was eventually hired, while #2 (right) was screened

1. Primary nodes (candidate and job description) are characterized by 18-dimensional feature vectors $\mathbf{x}_{q,\{i,j\}} \in \mathbb{R}^{18}$ derived from standardized psychometric assessments.
2. Entity nodes employ a composite feature representation obtained through multiple pooling operations on their respective embeddings:

$$\mathbf{x}_{e,\{i,j\}} = \left[\mathrm{mean}(\{\{\mathbf{v}_{e,\{i,j\}}\}\}), \mathrm{sum}(\{\{\mathbf{v}_{e,\{i,j\}}\}\}), \mathrm{max}(\{\{\mathbf{v}_{e,\{i,j\}}\}\})\right] \tag{2}$$

While entity semantics are partially encoded in the graph topology, their vectorial representations enable integration with psychometric data through GNN operations. This composition can be interpreted as an instance of DeepSets [43], allowing the model to learn permutation-invariant representations of variable-sized entity sets. Alternative approaches, such as Generative Topographic Mapping [11], could offer probabilistic entity representations, presenting an avenue for future investigation. In addition, we experimented with a naive one-hot encoding of the entity's features, which yielded inferior results.

4.4 Dataset Characteristics

The dataset encompasses $N = 9532$ graph instances, comprising $|V| = 133,448$ nodes and $|E| = 169,213$ edges in total. The edge distribution exhibits moderate variability, with a mean of 17.75 edges per graph ($\sigma = 1.43$) and a range of [15, 21]. This variation in edge density reflects the heterogeneous nature of candidate-job relationships captured by our similarity metrics.

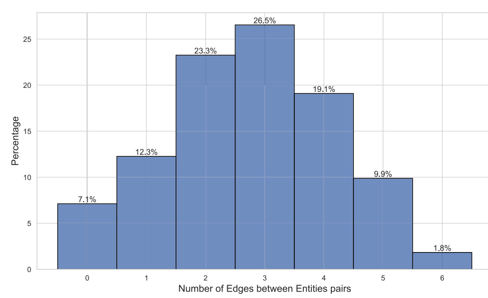
Statistical analysis reveals that 92.88% of graphs (8853 instances) contain at least one inter-entity edge, indicating substantial semantic overlap between candidate profiles and job requirements. The distribution of these connections provides discriminative structural features for the learning algorithm, as evidenced by the distinct edge patterns observed across different recruitment outcomes (see Fig. 3).

The edge count statistics presented in Fig. 3(right) reveal a noteworthy pattern: there is a statistical difference in the average number of edges across different labels. Graphs associated with higher label values (shortlisted and hired candidates) tend to have higher average edge counts (3.16 and 3.13, respectively) compared to rejected candidates (2.74), suggesting that our graph structure effectively captures relevant matching signals that GNNs are particularly well-suited to exploit.

It is compelling to observe the overall distribution of entities $\mathcal{E}$ and their respective edges among the labels. To visualize these patterns, we normalize their count over all graphs and highlight their difference w.r.t. the mean count in Fig. 4. The heatmap of edges ratios reveals intriguing patterns across different labels y and entity categories $\mathcal{E}$. Notably, the *Hard Skill* category consistently exhibits the highest edge ratios across all labels, suggesting its central role in connecting candidates to job vacancies. Interestingly, the `Hired` label ($y = 3$) shows the highest ratio for this category at 79.37%, indicating that the combination of hard skills is particularly crucial for successful hires. The *Role* category also shows high edge ratios, especially for shortlisted ($y = 2$) and hired candidates, emphasizing its importance in the later stages of the hiring process. In contrast, the *Education* and `Industry Sector` categories generally show lower edge ratios, with some variation across labels. `Industry Sector` and *Field of Education* are more prevalent for $y = 2$ than for $y = 3$ as desired. This could mean that those qualities are biasing factors for being shortlisted, suggesting that these two labels could also be merged into one. Finally, the *Soft Skill* category shows relatively consistent ratios across all labels, implying its consistent importance throughout the hiring process.

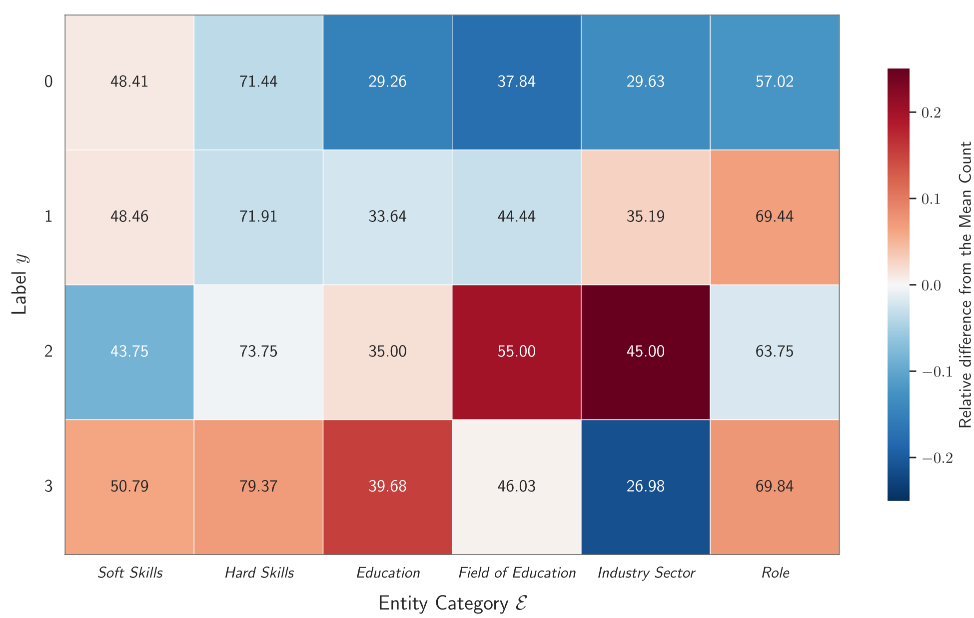
All the above desired properties are the outcome of setting $k = 10, p = 4$, as different hyperparameters lead to worse conditions (fewer edges, skewed distributions, minor correlations).

Fig. 3 Analysis of inter-entity edges across the CVs and JDs. Distribution of edge counts (right), and statistical summary grouped by graph label (left)



Label y	Mean	Std. Dev.
0	2.74	1.43
1	3.03	1.44
2	3.16	1.55
3	3.13	1.36

Fig. 4 Distribution of edge ratios across labels and entity categories $\mathcal{E}$. The heatmap displays the relative difference from the mean (normalized) count for each category. Colors range from dark blue (significantly below mean) through white (at mean) to dark red (significantly above mean). Numerical values represent the original edge ratios (% of cases whether there is a link between those entities in the bipartite CJM graphs)



5 Proposed Model

We formulate a specialized GNN architecture tailored for candidate-job matching graphs, systematically exploring diverse graph convolution operators and dual learning objectives. Our architecture comprises several key components engineered to process the heterogeneous nature of recruitment data, leveraging both structural information encoded in graph topology and the rich node features derived from textual entity embeddings and psychometric assessment data. Figure 5 presents a schematic illustration of the complete processing pipeline.

5.1 Learning Tasks and Loss Functions

Prior to detailing the architectural specifications, we formulate the CJM problem through two distinct learning paradigms: ordinal and binary graph classification. These complementary approaches capture different aspects of the recruitment decision process.

5.1.1 Ordinal Classification

Given the inherently sequential nature of recruitment stages (`Rejected` → `Interviewed` → `Shortlisted` → `Hired`), we implement ordinal classification using a Consistent Rank Logits (CORAL) output layer [6]. The CORAL framework models ordinal relationships by learning a series of binary classifiers for the output layer, each corresponding to a threshold between consecutive ranks. The loss function for this task is formulated as:

$$\mathcal{L}_{\text{ordinal}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K-1}\left[y_g^k \log(\sigma(f_k(\mathbf{x}_i))) + (1-y_g^k)\log(1-\sigma(f_k(\mathbf{x}_i)))\right] \tag{3}$$

where $K = 4$ is the number of ordinal classes, y_g^k denotes the binary indicator for the g-th graph sample at the k-th threshold, $f_k(\mathbf{x}_g)$ represents the model output for the k-th threshold, and $\sigma(\cdot)$ is the sigmoid activation function.

5.1.2 Binary Classification

As an alternative formulation, we aggregate positive outcomes ($y^* = \{\texttt{Interviewed}, \texttt{Shortlisted}, \texttt{Hired}\}$), creating a binary distinction between rejected candidates ($y = 0$) and those who advanced in the recruitment process ($y = y^*$). The binary classification loss function is expressed as:

$$\mathcal{L}_{\text{binary}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_g \log(\tilde{y}_g) + (1-y_g)\log(1-\tilde{y}_g)\right] \tag{4}$$

where y_g denotes the binary label (0 or $1 = y^*$) and $\tilde{y}_g$ represents the model's prediction for the g-th graph sample. To counteract the significant class imbalance, we apply a weighting factor $w = 10$ to the positive class instances (y^*) in both loss formulations, which empirically improves performance metrics.

From a recruitment and business perspective, our dual approach to the learning task offers distinct analytical advantages. The ordinal classification aligns with the sequential nature of recruitment pipelines, providing insights into factors contributing to candidate advancement at each distinct stage. Conversely, binary classification presents a more efficient approach for large-scale preliminary candidate evaluation, enabling rapid identification of promising candidates worthy of further consideration.

5.2 Model Architecture

Our architecture comprises three principal components: (1) a pre-aggregation module for heterogeneous feature processing, (2) multiple graph convolutional layers for message propagation culminating in a pooling operation, and (3) a classification head for final predictions.

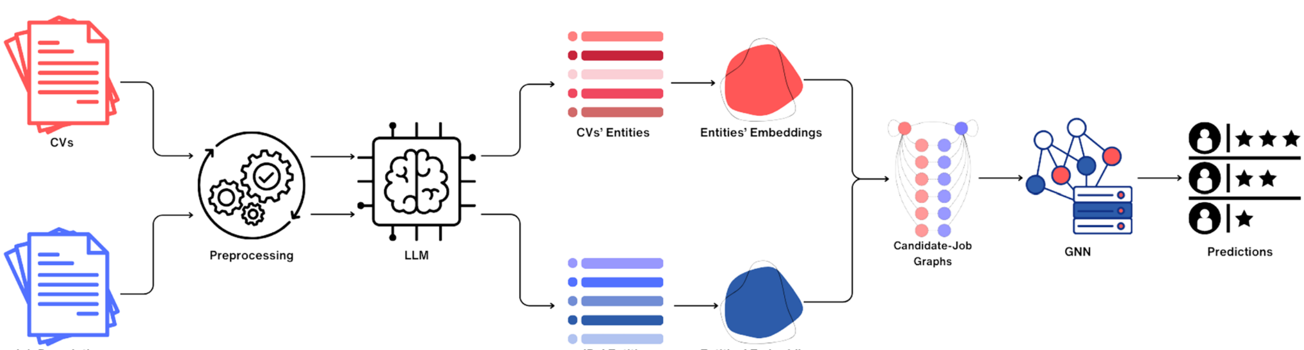


Fig. 5 Data pipeline of the proposed model for CJM. Input data (CV and JDs) are preprocessed, an LLM extracts relevant entities which are translated into embeddings. These are used to build CJM graphs, which are employed to train a GNN

5.2.1 Pre-aggregation Layer

The initial component processes heterogeneous node features through specialized linear projections. We implement seven independent linear transformations that map features to a common hidden dimension: one dedicated to the primary nodes (candidate and job description) processing the psychometric features $\mathbf{x}_{q,\{i,j\}}$, and six for the entity nodes processing their respective aggregated embeddings $\mathbf{x}_{\epsilon,\{i,j\}}$. This transformation ensures dimensional consistency and semantic alignment prior to the message-passing layers.

5.2.2 Graph Convolutional Layers

The core of our architecture consists of parameterized graph convolutional layers. We systematically evaluate five distinct graph convolution operators, based on their effectiveness on a broad spectrum of applications:

- *Graph Convolutional Network (GCN)* [18] The standard graph convolution operator that aggregates neighborhood information through normalized adjacency matrices.
- *Multiplicative Integration GNN (MIGNN)* [13] An adaptation that employs multiplicative integration of node features, enabling the model to capture higher-order interactions between nodes' attributes.
- *Graph Isomorphism Network (GIN)* [39] A theoretically powerful architecture that can distinguish different graph structures through injective neighborhood aggregation.
- *Graph Attention Network (GAT)* [36] Incorporates attention mechanisms to weigh the importance of different neighbors dynamically.
- *GraphConv* [8] A general formulation combining local and global graph properties through separate weight matrices.

Each convolutional operation is followed by *Graph Normalization* [7] to stabilize training dynamics and a *LeakyReLU* non-linearity to introduce representational capacity. We additionally implement optional Jumping Knowledge connections [38], enabling adaptive fusion of multi-scale node representations across different layers of the network hierarchy.

5.2.3 Model Output and Readout

The terminal component implements a hierarchical readout phase that transforms node-level representations into graph-level predictions. This process entails:

1. Global graph pooling operations (sum, mean, and max pooling) applied to node embeddings (similarly to Item 2)
2. A sequence of fully-connected layers with non-linear activations (also called *deep readout*) to extract graph-level representations.
3. Task-specific output projection aligned with either the ordinal or binary learning objective.

5.3 Model Configuration

The model's performance is governed by several critical hyperparameters, including the network depth (number of message-passing layers), representational capacity (hidden channel dimensions), regularization strength (dropout rate), and architectural complexity (number of readout layers). The complete hyperparameter search space is enumerated in Table 4, with all unspecified parameters maintained at their default values according to the PyTorch Geometric implementation.

6 Experimental Evaluation

To rigorously assess the efficacy of our graph-based approach to CJM, we conducted a comprehensive empirical evaluation comparing various model architectures and learning paradigms. Our experimental protocol was designed to evaluate the models' discriminative capabilities in identifying promising candidates from large applicant pools—a scenario that directly addresses the practical needs of HR practitioners.

6.1 Implementation Details

We implemented all models using `PyTorch 2.4` and `PyTorch Geometric` [10], with training conducted on computation nodes equipped with NVIDIA RTX A5000 GPUs. Hyperparameter optimization was performed via Bayesian optimization using the Optuna framework [3], executing 100 trials per configuration to thoroughly explore the parameter space. Each trial comprised 300 training epochs with an early stopping mechanism (patience = 50) based on validation loss to prevent overfitting while ensuring

Table 4 Hyperparameters grid for the GNN model in candidate-job vacancy matching

Hyperparameters	Range/values
Hidden channels	32 to 256 (step 32)
Number of GC Layers	$\{1, 2, 3, 4, 5\}$
Learning rate	$\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$
Dropout rate	$\{0.0, 0.5\}$
Jumping knowledge (JK)	{`True`, `False`}
Deep readout layers	$\{0, 1, 2, 3\}$

convergence. All models were trained end-to-end using the *AdamW* optimizer [25] with weight decay ($\lambda = 1e^{-5}$) and a cosine annealing learning rate schedule. The optimization objective was formulated to minimize the loss on the validation set.

Reproducibility Statement Due to the sensitive nature of the recruitment data used in this study, which contains personal and proprietary information, we cannot publicly release the dataset. However, to facilitate reproducibility and further research, we will make our code, model implementations, and graph construction methodology available upon reasonable request to qualified researchers who agree to comply with privacy protection protocols. The paper provides comprehensive details on the graph construction process, entity extraction methodology, and model architectures that should enable researchers to replicate our approach on other HR datasets.

6.2 Dataset Partitioning Strategy

Given the inductive nature of our graph classification approach, we implemented a stratified partitioning strategy that preserves the integrity of complete recruitment scenarios. Our dataset comprises $S = 62$ distinct selection processes, each associated with its corresponding set of CJM graphs. We partitioned these processes into training (70%), validation (20%), and test (10%) sets at the process level rather than at the individual graph level.

This process-level stratification serves two critical purposes: (1) it maintains the validity of each recruitment scenario by ensuring that all graphs pertaining to a specific recruitment process are allocated to the same partition, and (2) it simulates authentic deployment conditions where models must generalize to entirely novel recruitment scenarios without prior exposure to their specific characteristics or requirements.

To enhance statistical robustness and mitigate potential sampling bias, we generated five distinct partitioning schemes using different random seeds while maintaining the consistent 70-20−10 ratio across splits.

In parallel, we created a filtered version of the dataset that more accurately reflects the operational recruitment pipeline of our industrial partner. This filtered variant retains only 10% of candidates lacking questionnaire data (randomly sampled), while preserving all candidates who completed the psychometric assessment. Although this filtering reduces the overall dataset size, it provides a more representative sample of the actual candidate evaluation process, particularly emphasizing candidates who engaged more comprehensively with the assessment methodology—a criterion that recruiters consider essential for evaluating person-environment fit [16, 23].

6.3 Baseline Architecture and Evaluation Metrics

To establish a rigorous comparative benchmark, we implemented a Multi-Layer Perceptron (MLP) baseline that operates within the same experimental framework as our GNN models. This architecture begins with identical pre-aggregation processing of node features from the CJM graphs, but substitutes graph convolution operations with a parameterized deep readout phase comprising multiple fully-connected layers (ranging from 1 to 5). This controlled comparison isolates the specific contribution of graph structural information to model performance, enabling direct assessment of the hypothesized benefits of graph-based learning in the CJM context.

Given the pronounced class imbalance inherent in our dataset, we resorted to *balanced* accuracy, *weighted* F1 score. Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are included to provide insight into the magnitude of prediction errors, particularly relevant for the ordinal task. These metrics are computed on the test set for each of the five dataset splits, and we report both mean values and standard deviations to ensure a robust evaluation.

7 Results and Analysis

We present a comprehensive empirical evaluation of our graph-based CJM approach across multiple model architectures and learning paradigms. Tables 5 and 6 summarize performance metrics for both complete and filtered datasets under binary and ordinal learning objectives.

Ordinal Classification Task Our experimental results reveal a consistent performance plateau across all architectures in the ordinal classification task, with balanced accuracy converging at 25% for both dataset variants. This performance is statistically equivalent to a prior classifier that consistently predicts the majority class ($y = 0$). The uniform performance across architectures-spanning from the MLP baseline to sophisticated GNN variants-suggests a fundamental limitation in the ordinal formulation of the CJM problem rather than deficiencies in specific model architectures.

The high weighted F1 scores ($\approx 92.9\%$) observed alongside low balanced accuracy presents an apparent contradiction that merits closer examination. This discrepancy stems from the severe class imbalance in our dataset (see Table 2), where the weighted F1 metric heavily favors performance on the dominant rejection class. The consistently low MAE values ($\approx$0.068) indicate that prediction errors typically occur between adjacent ordinal categories rather

Table 5 Comparison of losses across different datasets and tasks for different models

Dataset	Task	Model	Train loss	Validation loss	Test loss
Filtered	Ordinal	MLP	$323.7_{\pm 36.2}$	$115.2_{\pm 15.0}$	$67.5_{\pm 10.3}$
		GCN	$307.3_{\pm 25.5}$	$112.3_{\pm 14.3}$	$65.0_{\pm 10.9}$
		MIGNN	$310.5_{\pm 20.8}$	$108.9_{\pm 15.3}$	$63.4_{\pm 9.6}$
		GIN	$325.3_{\pm 26.0}$	$112.0_{\pm 13.9}$	$65.0_{\pm 11.6}$
		GAT	$321.2_{\pm 19.6}$	$109.7_{\pm 16.0}$	$62.7_{\pm 9.1}$
		GraphConv	$316.6_{\pm 34.7}$	$110.8_{\pm 14.8}$	$65.0_{\pm 10.4}$
	Binary	MLP	$131.6_{\pm 18.3}$	$52.4_{\pm 5.5}$	$31.5_{\pm 5.0}$
		GCN	$109.7_{\pm 11.0}$	$50.3_{\pm 5.4}$	$30.5_{\pm 4.1}$
		MIGNN	$101.1_{\pm 6.4}$	$47.7_{\pm 5.3}$	$30.0_{\pm 4.2}$
		GIN	$111.9_{\pm 9.3}$	$50.2_{\pm 6.7}$	$29.8_{\pm 4.3}$
		GAT	$127.1_{\pm 6.9}$	$50.0_{\pm 7.2}$	$30.3_{\pm 5.1}$
		GraphConv	$102.4_{\pm 7.4}$	$49.0_{\pm 7.5}$	$29.7_{\pm 5.5}$
Complete	Ordinal	MLP	$389.3_{\pm 49.0}$	$112.2_{\pm 14.2}$	$65.9_{\pm 8.2}$
		GCN	$343.2_{\pm 29.6}$	$103.0_{\pm 17.0}$	$64.4_{\pm 11.7}$
		MIGNN	$402.4_{\pm 19.3}$	$107.5_{\pm 16.2}$	$60.3_{\pm 11.2}$
		GIN	$418.4_{\pm 28.1}$	$111.7_{\pm 15.4}$	$63.2_{\pm 10.0}$
		GAT	$430.0_{\pm 57.2}$	$109.1_{\pm 15.6}$	$62.1_{\pm 8.8}$
		GraphConv	$436.5_{\pm 65.5}$	$109.3_{\pm 16.0}$	$62.2_{\pm 9.0}$
	Binary	MLP	$180.5_{\pm 26.0}$	$50.4_{\pm 5.8}$	$30.4_{\pm 4.5}$
		GCN	$140.3_{\pm 17.0}$	$45.0_{\pm 7.8}$	$28.5_{\pm 3.7}$
		MIGNN	$146.5_{\pm 13.2}$	$48.3_{\pm 6.8}$	$29.1_{\pm 4.0}$
		GIN	$181.7_{\pm 18.1}$	$50.2_{\pm 7.0}$	$29.0_{\pm 4.3}$
		GAT	$190.2_{\pm 9.8}$	$49.7_{\pm 6.8}$	$29.7_{\pm 3.0}$
		GraphConv	$185.6_{\pm 13.5}$	$49.4_{\pm 6.2}$	$29.8_{\pm 4.2}$

than across the entire ordinal spectrum. This finding suggests that while models struggle with precise ordinal discrimination, they maintain a degree of ordinal awareness in their misclassifications.

Binary Classification Task The binary classification paradigm demonstrates substantially more promising results, with all architectures achieving balanced accuracies significantly above the 50% random baseline. In the filtered dataset, the GCN architecture demonstrates superior discriminative capability (61.8% balanced accuracy), followed by GAT (60.5%) and GraphConv (59.1%). The performance differential between graph-based methods and the MLP baseline (50.8%) provides compelling evidence for the efficacy of structural information in CJM tasks.

For the complete dataset, which encompasses a broader spectrum of candidates including those without psychometric assessments, we observe improved performance across all architectures. The GCN model maintains its superior position with 65.4% balanced accuracy, followed by MIGNN (62.2%) and GraphConv (58.9%). This performance enhancement with increased data volume, despite the inclusion of candidates with imputed psychometric features, suggests that the models effectively leverage the graph structure to compensate for missing feature information.

The MAE values for binary classification (ranging from 0.107 to 0.268) exceed those of the ordinal task, which is an expected consequence of the binary formulation—any misclassification incurs a full unit of error. However, these values must be interpreted in conjunction with the balanced accuracy metrics, which reveal the models' improved capacity to discriminate between positive and negative outcomes in the binary setting.

Discriminative Capabilities Across Architectures Analysis of the normalized confusion matrices in Fig. 6 reveals distinctive patterns in how each architecture negotiates the extreme class imbalance—945 samples in class 0 versus 47 in class 1. The MLP baseline achieves high accuracy on the majority class (92.80%) but demonstrates critically poor performance on the minority class, correctly identifying only 8.51% of positive cases. This represents a significant limitation for practical recruitment scenarios where identifying promising candidates is the primary objective.

In contrast, graph-based architectures demonstrate improved minority class detection capabilities. The GCN architecture achieves the highest minority class sensitivity at 48.94%, albeit with reduced majority class accuracy (74.39%). This trade-off pattern is consistent across all graph models, with each establishing a distinct operating point along the sensitivity-specificity frontier. Notably, MIGNN offers a balanced compromise with 84.55% majority class accuracy and 29.17% minority class sensitivity, potentially representing an optimal configuration for scenarios where both precision and recall are valued.

Interestingly, the Complete dataset facilitates improved minority class detection across all architectures, suggesting that the additional training samples, despite being predominantly from the majority class, contribute positively to the learning dynamics. This counterintuitive finding may reflect the models' increased capacity to identify subtle patterns of matching when exposed to a more diverse candidate pool.

Graph Convolution Mechanism Analysis The consistent outperformance of graph-based models validates our hypothesis that graph structures effectively encode meaningful semantic relationships in recruitment data. While each architecture shows distinct strengths, GCN emerges as the most reliable across our experiments, particularly excelling in balanced accuracy. This suggests that the straightforward spectral convolution mechanism, which uniformly aggregates neighborhood information through normalized adjacency matrices, effectively captures the essential matching patterns in our CJM graphs.

The attention mechanism in GAT and the multiplicative interactions in MIGNN also demonstrate efficacy, particularly in handling class imbalance scenarios. Their capacity to adaptively weight node contributions enables more

Table 6 Comparison of models performance metrics (Balanced Accuracy (%), Weighted F1, MAE, RMSE) on the test set across different datasets and tasks

Dataset	Task	Model	B. Accuracy	W. F1	MAE	RMSE
Filtered	Ordinal	MLP	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		GCN	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		MIGNN	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		GIN	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		GAT	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		GraphConv	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
	Binary	MLP	$50.8_{\pm 2.3}$	$89.5_{\pm 6.1}$	$0.107_{\pm 0.105}$	$0.298_{\pm 0.134}$
		GCN	$61.8_{\pm 1.8}$	$80.5_{\pm 5.6}$	$0.268_{\pm 0.082}$	$0.511_{\pm 0.082}$
		MIGNN	$56.7_{\pm 3.8}$	$86.1_{\pm 2.5}$	$0.184_{\pm 0.045}$	$0.426_{\pm 0.050}$
		GIN	$58.6_{\pm 2.3}$	$85.5_{\pm 1.9}$	$0.194_{\pm 0.032}$	$0.439_{\pm 0.038}$
		GAT	$60.5_{\pm 3.9}$	$81.5_{\pm 7.3}$	$0.251_{\pm 0.103}$	$0.492_{\pm 0.096}$
		GraphConv	$59.1_{\pm 3.5}$	$83.5_{\pm 4.0}$	$0.225_{\pm 0.063}$	$0.469_{\pm 0.070}$
Complete	Ordinal	MLP	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		GCN	$25.1_{\pm 0.2}$	$92.8_{\pm 1.1}$	$0.072_{\pm 0.014}$	$0.353_{\pm 0.024}$
		MIGNN	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.347_{\pm 0.018}$
		GIN	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		GAT	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.348_{\pm 0.017}$
		GraphConv	$25.0_{\pm 0.0}$	$92.9_{\pm 0.8}$	$0.068_{\pm 0.007}$	$0.346_{\pm 0.016}$
	Binary	MLP	$55.0_{\pm 4.8}$	$85.0_{\pm 6.9}$	$0.191_{\pm 0.114}$	$0.412_{\pm 0.146}$
		GCN	$65.4_{\pm 3.6}$	$87.0_{\pm 1.8}$	$0.173_{\pm 0.029}$	$0.414_{\pm 0.036}$
		MIGNN	$62.2_{\pm 1.4}$	$86.7_{\pm 2.0}$	$0.176_{\pm 0.034}$	$0.418_{\pm 0.044}$
		GIN	$58.1_{\pm 4.2}$	$85.3_{\pm 4.9}$	$0.191_{\pm 0.088}$	$0.422_{\pm 0.116}$
		GAT	$56.9_{\pm 2.8}$	$87.3_{\pm 4.2}$	$0.159_{\pm 0.074}$	$0.387_{\pm 0.097}$
		GraphConv	$58.9_{\pm 3.7}$	$84.8_{\pm 2.2}$	$0.206_{\pm 0.035}$	$0.452_{\pm 0.038}$

The values averaged over the test set across five runs have rounded standard deviations to the nearest decimal point

nuanced feature integration, resulting in improved minority class detection. The relatively weaker performance of GIN, despite its theoretical advantages in graph isomorphism testing, suggests that the CJM task may not heavily depend on fine-grained structural discrimination but rather on effective feature propagation along established semantic pathways.

These findings hold significant practical implications for recruitment scenarios. The graph-based approach enables recruiters to prioritize manual assessment of candidates classified as promising by the model, striking a reasonable balance between efficient evaluation of large applicant pools and identification of high-potential candidates.

7.1 Limitations and Avenues for Improvement

Our research provides notable insights into using GNNs for CJM, but it is also crucial to acknowledge its limitations and seek ways to progress.

One of the most striking observations from our results is the poor performance of all models in the ordinal task. Our architectures perform no better than a simple prior classifier, indicating that our current approach to ordinal classification in this context is suboptimal. The very nature of treating CJM as an ordinal classification problem may be flawed, as the assumption of clear, ordered levels of suitability between candidates and job positions might be an oversimplification of a complex, multidimensional problem. To address this issue, we could consider reformulating the problem as a multi-class classification task, where each suitability level is treated as a separate class without an inherent order. This approach would allow models to learn more nuanced relationships between features and outcomes without imposing an ordinal scale. Additionally, exploring regression-based approaches or even multi-task learning, where we predict multiple aspects of CJM or a matching preference, could equally support HR recruiters.

Another potential limitation lies in the features used to represent candidates and jobs. Important aspects of a candidate's suitability, such as soft skills, cultural fit, ambitions, or specific experiences that are not easily quantified, are missing from our data based only on CVs and assessment questionnaires. To improve this, we could incorporate more detailed candidate and job information, such as interview transcripts or data on the hiring company. Additional pertinent information, including years of professional experience, references, salary, and geographical location, are also

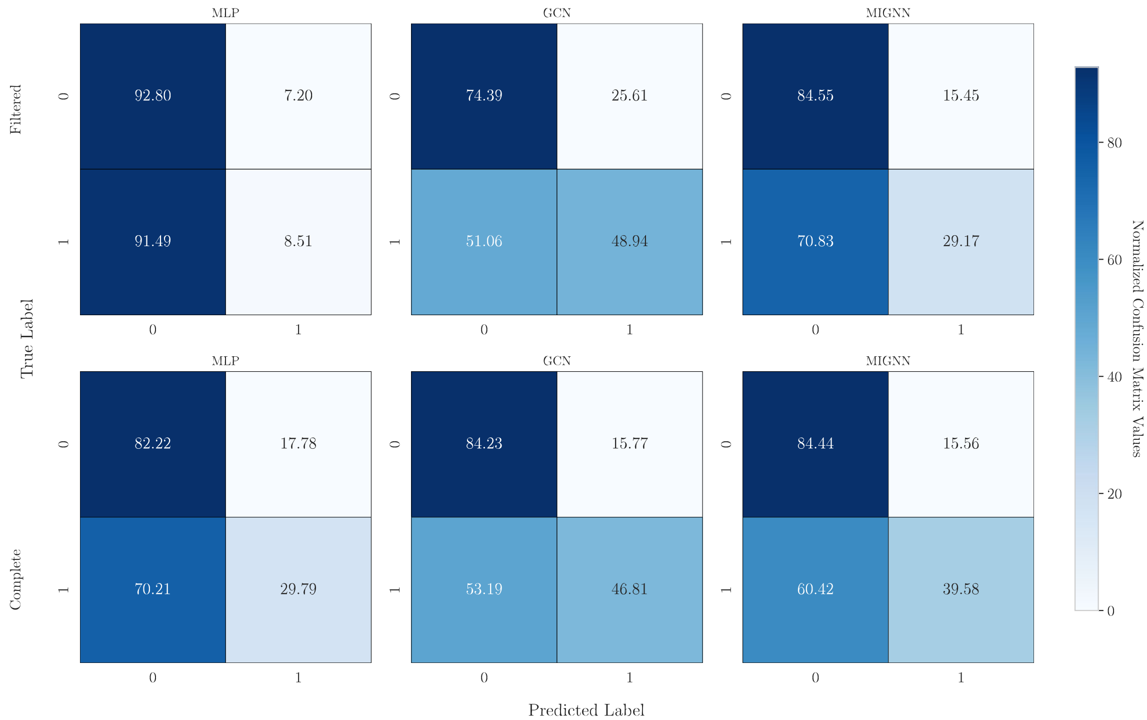


Fig. 6 Comparison of confusion matrices for six different models in the CJM task. Each cell displays the normalized percentage (over true labels) averaged over the test splits. The class distribution is divided into 945 samples for class 0 and 47 in class 1. The color intensity indicates the proportion of samples in each true class that were predicted as each possible class, with darker blues representing higher proportions

considered for the recruitment process. Nevertheless, these have been excluded from our analysis due to privacy and ethical concerns, possibly hindering the models' effectiveness. In general, more HR data needs to be collected to increase the pool of candidates and selections and consequently extend this analysis, yet this is a costly and time-consuming task.

The graph structure of our data presents another area for potential improvement. The graph construction methodology, while effective, currently relies on heuristic similarity measures derived from embedding proximity. This approach, though theoretically motivated, may not optimally capture domain-specific nuances in recruitment matching. Future iterations could explore different graph construction methods, incorporating more domain knowledge to create edges that better represent meaningful relationships between candidates and jobs, or by letting the model learn the optimal graph topology.

Finally, the extreme class imbalance inherent in recruitment datasets poses ongoing challenges for model optimization. We adopted a loss weighting strategies, but more sophisticated approaches such as synthetic minority oversampling techniques (SMOTE) for graph data or hierarchical classification frameworks might further improve performance on the critical minority class detection task.

8 Conclusion

This investigation into graph neural networks for candidate-job matching demonstrates the potential of graph-based approaches to augment human decision-making in recruitment processes. Our work makes several noteworthy contributions to the intersection of graph representation learning and human resource analytics:

1. *Integration of LLMs and GNNs* We establish an effective methodological pipeline that leverages the semantic extraction capabilities of Large Language Models to generate structurally meaningful graph representations, which are then processed through specialized graph neural architectures.
2. *Inductive Graph Representation Framework* We introduce an inductive graph learning paradigm for CJM that constructs purpose-built graphs for each candidate-job pairing. Compared to other methods in the literature, this approach allows for a more nuanced capture of the complex relationships between candidates' attributes and job requirements.
3. *Empirical Validation of Graph Structural Information* Through rigorous comparative analysis across multiple

architectures, we demonstrate that graph-based methods consistently outperform equivalent non-graph alternatives in binary classification tasks. This provides empirical evidence for the value of structural information in recruitment matching beyond what can be captured through feature vectors alone.

4. *Real-World Application* We addressed the challenges of utilizing AI in a real-world HR setting by collaborating with HR recruiters and processing actual CVs. This resulted in a graph-based AI model that could effectively support recruiters in screening and assessing new candidates.
5. *Practical Deployment Considerations* By evaluating performance under realistic class imbalance scenarios and across multiple model configurations, we provide actionable insights for practitioners implementing graph-based recruitment support systems.

While our current results show promise, they also reveal the complexity of HR data and the CJM problem. The performance gap between our models and ideal predictors underscores the need for continued research and development in this area, particularly for trustworthy and production-ready applications.

Looking forward, several promising research directions emerge from our findings. The integration of organizational context and cultural dimensions into the graph construction process could enhance matching precision beyond skill-based compatibility. Advanced techniques for handling extreme class imbalance in graph data could further improve minority class detection capabilities. Additionally, interactive systems that combine model recommendations with recruiter expertise could leverage the complementary strengths of human and artificial intelligence.

In conclusion, our research demonstrates that graph neural networks offer a promising approach for modeling the complex, relational nature of candidate-job matching. However, their effective deployment requires careful consideration of the inherent limitations and ethical implications of automated recruitment systems. The path forward lies in developing hybrid approaches that augment rather than replace human judgment, leveraging the pattern recognition capabilities of graph-based models while preserving the contextual understanding and ethical oversight that human recruiters provide. By maintaining this balance, graph-based recruitment systems can enhance efficiency and fairness in talent acquisition while respecting the fundamentally human dimensions of employment decisions.

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Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by PF and MH. The first draft of the manuscript was written by PF and all authors commented on previous versions of themanuscript. All authors read and approved the final manuscript.

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Data Availibility Due to the sensitive nature of the data, the dataset is not publicly available.

Code Availability Code will be available upon request.

Declarations

Conflict of interest Author FF is a member of the board of directors of Amajor and head of its R&D Department. Author AS declares they have no financial interests.

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